MAT347: Groups, Rings, and Fields

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Motivation and First Definitions

Some general things:

1. Instructor: Florian Herzig
2. Office Hours: TBA
3. TAs: Nischay Reddy, Aleksandr Popkovich
4. Textbook: Abstract algebra by Dummit and Foote (Wiley, 3rd ed.)

There will be a quiz, every two weeks in tutorial, otherwise there will be some worksheets and discussion (No quiz tomorrow!), but there will be one next week, it will be like a mini test worth a small part of the grade, in order to receive feedback. There will be assignments every two weeks, these are worth ε of our grade considering how easy it is for one to use their favorite LLM and generate solutions. There will 3 term tests on the following dates:

1. November 5
2. January 21
3. March 11

and a final exam in April, to be announced. If you do not spend atleast 5-10 hours outside of class on this course it will be hard to succeed. There are more information in the course website (to add later because). There will be some problems posted regularly to give us an opportunity to practice
Now one might be wondering what abstract algebra is about, some of the highlights of this course will appear in the second half of this course, such as

1. Why we can construct regular n-gons with a straight edge and compass for some n? Some interesting ones include $n = 3, 4, 5, 16, 17$ but not $n = 7$
2. Why is there a formula for solving, quadratic, cubic, quartic equations, but not quintic equations? This led to the development to Galois theory! (Galois was a young mathematician who died in a duel during the early 1830s)

The first thing we will discuss is group theory, this area has been in more recent development than other topics encountered in previous undergrad courses, the modern definition of a group dates back to the 1880s. In a nutshell, group theory is the study of symmetries, this is relevant in physics, geometry, and even in analysis (Fourier theory studies the symmetries on S^1) and other fields. More concretely, if you are interested in puzzles, you will encounter lots of them here.

It took lots of work for mathematicians to develop the first example of a group, which are the permutations which we encounter in first year.

Example. The symmetries of n identical objects are nothing but permutations, which are bijections

$$\sigma : \{1, \dots, n\} \to \{1, \dots, n\}.$$

We will denote these by S_n.

For example, when $n = 4$, we have

$$\begin{array}{cccc} \sigma_1 & & & \\ 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 4 \end{array} \qquad \begin{array}{cccc} \sigma_2 & & & \\ 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 1 & 3 & 4 \end{array} \qquad \begin{array}{cccc} \sigma_3 & & & \\ 1 & 2 & 3 & 4 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 2 & 3 & 4 & 1 \end{array}$$

and so on. In total, there are

$$|S_4| = 4! = 24$$

such permutations. There are basic operations that we can perform, such as composition

$$S_n \times S_n \to S_n, \qquad (\sigma, \tau) \mapsto \sigma \circ \tau,$$

and inversion

$$S_n \to S_n, \qquad \sigma \mapsto \sigma^{-1}.$$

These are the most fundamental ones.

These operations satisfy the following properties

1. Associativity $f \circ (g \circ h) = (f \circ g) \circ h$.
2. There is an identity element e which satisfies $e \circ f = f = f \circ e$ for $f \in S_n$.
3. There is an inverse f^{-1} $f \circ f^{-1} = e = f^{-1} \circ f$

Example. The rotational symmetries of a cube = Rotation of $\mathbb{R}^3$ preserve the cube (rotation about a space diagonal). There are 24 symmetries, in fact, these are the same symmetries as S_4.

One of the small things group theorists say, take a number n and try to classify things, for some n its easy, and for some it is really hard. The same thing happens in geometry where we attempt to classify manifolds.

Definition. A binary operation $*$ on a set X is a function $X \times X \to X$

Example.

1. Addition
2. Multiplication
3. Division (on $\mathbb{R} \setminus \{0\}$)

Definition. A group is a set G together with a binary operation $*$ such that

1. Associativity $(a * b) * c = a * (b * c)$ for all $a, b, c \in G$
2. $\exists e \in G$ such that $e * g = g = g * e$ for all $g \in G$
3. For each $g \in G$ there exists $g^{-1} \in G$ such that $g * g^{-1} = e = g^{-1} * g$

Remark: Often, we say that "G is a group" and we will say that if it is clear what the operation is.

Example. Some examples of groups include:

1. $(S_n, \circ)$
2. Any number field with addition
3. $k \setminus \{0\}$ where k is any field under $\times$
4. $(\mathrm{GL}_n(\mathbb{R}), \times)$
5. $\{e\}$
6. $(\mathbb{R} \setminus \{0\})$ with $\times$
7. Any vector space over any field.

Example. Some non examples include:

1. $(\mathbb{R}, \times)$
2. $(\mathbb{Z}, \times)$
3. $(\mathbb{N}, +)$

And many more.

Definition. We say that a group G is abelian or commutative if $x * y = y * x$

Exercise: find 2 elements in S_3 that do not commute.

Proposition. Let G be a group, the following properties hold:

1. e is unique
2. Inverses are unique
3. $(x^{-1})^{-1} = x$
4. $(x * y)^{-1} = y^{-1} * x^{-1}$
5. $x_1 * \cdots * x_n$ is independent of the use of brackets
6. Cancellation laws hold, if $x * y = x * z$ then $y = z$ and similarly if $y * z = y * x$ then $x = z$.

Proof. 1. Left as an exercise.

2. Suppose that x', x'' are inverses of $x \in G$. Then

$$\begin{aligned} (x' * x) * x'' &= x' * (x * x'') && \text{(Associativity)} \\ e * x'' &= x' && \text{(Existence of inverses)} \\ x'' &= x' && \text{(Existence of the identity)} \end{aligned}$$

3. Left as an exercise.
4. Left as an exercise

5. Apply strong induction (take $n \geq 3$)
6. Left as an exercise. (Hint: Multiply inverses)

□

Remark: $(x_1 * \cdots * x_n)^{-1} = x_n^{-1} * \cdots * x_1^{-1}$

Now, in order to encode the "structure" of a group $(G, *)$ (here G has to be small! $G = \{e, ..., g_n\}$), we can use a multiplication table:

$*$	e	g_2	$\cdots$	g_n
e	e	g_2	$\cdots$	g_n
g_2	g_2			
$\vdots$	$\vdots$		$x * y$	
g_n	g_n			

Example. If $|G| = 2$ and $G = \{e, g\}$, then

$*$	e	g
e	e	g
g	g	e

Definition. for $x \in G$, $n \in \mathbb{Z}$ we define

$$x^n = \begin{cases} \underbrace{x \cdots x}_{n} & \text{if } n > 0 \\ e & \text{if } n = 0 \\ \underbrace{x^{-1} \cdots x^{-1}}_{n} & \text{if } n < 0 \end{cases}$$

Proposition. For $x \in G$ and $n, m \in \mathbb{Z}$, we have

$$\begin{cases} x^n x^m = x^{m+n} \\ (x^m)^n = x^{mn} \end{cases}$$

Proof. (Hint: For (1) take cases depending on m, n. As for (2) it is the same idea but there will be fewer cases) □

Definition. Let G be a group and $a \in G$. We say that the order of the element a is the smallest positive integer n such that $a^n = 1$ if there is such an integer, otherwise we say that the order of a to be infinity.

Example. If $G = \mathbb{C} \setminus \{0\}$ (with $\times$), then $e^{\frac{2\pi i}{n}}$ has order n and 2 has order ∞.

Equivalence Relations and $\mathbb{Z}/n\mathbb{Z}$

Last time we discussed: Group theory! The corresponding sections for the week are now posted on the website. We talked about the history of the subject and how the symmetries of the cube motivated the first definition of a group, namely a set G with a binary operation $*$ such that the following are satisfied:

1. Associativity
2. The existence of an identity element
3. The existence of an inverse

Some of the examples that we looked at were

1. Symmetries of the cube
2. $(S_n, \circ)$
3. $(\mathbb{Z}, +)$

Recall that for $x \in G,\ n \in \mathbb{Z}$ we define

$$x^n = \begin{cases} \underbrace{x \cdots x}_{n} & \text{if } n > 0 \\ e & \text{if } n = 0 \\ \underbrace{x^{-1} \cdots x^{-1}}_{n} & \text{if } n < 0 \end{cases}$$

We also proved (by cases) that

$$\begin{cases} x^n x^m = x^{m+n} \\ (x^m)^n = x^{mn} \end{cases}$$

We also defined the order of an element $a \in G$ at the end of last lecture, which is the smallest positive integer n such that $a^n = 1$ if there is such an integer, otherwise we say that the order of a is infinity. Today we will discuss the group $\mathbb{Z}/n\mathbb{Z}$. Fix $n \geq 1$, for $a, b \in \mathbb{Z}$, recall that $a \equiv b \bmod n$ or $a \equiv_n b$ if $n|(a-b)$.

Definition. A relation $\sim$ on X is a subset $R_\sim$ of $X \times X$. We write $x \sim y$ if $(x, y) \in R_\sim$

Example. $\leq$, $\leq$, divisibility

Definition. We say that:

1. $\sim$ is reflexive if $x \sim x$ for all $x \in X$
2. $\sim$ is symmetric if $x \sim y$ then $y \sim x$ for all $x, y \in X$.
3. $\sim$ is transitive if $x \sim y$ and $y \sim z$ then $x \sim z$ for all $x, y, z \in X$.

If $\sim$ satisfies these properties, then $\sim$ is an equivalence relation.

Example. Some examples include $=$ on any set X, $\equiv$ on $\mathbb{Z}$.

Definition. For $x \in X$, we define the equivalence class of x

$$[x] = \{y \in X \ : \ y \sim x\}$$

Equivalence classes form a partition of X, we denote this by

$$X = \bigcup_{x \in X} [x]$$

and for all $x, y \in X$, we have that $[x] = [y]$ or $[x] \cap [y] = \varnothing$.

Definition. Let X be a set and $\sim$ be an equivalence relation. We define the set

$$X/\sim := \{[x] : x \in X\}$$

and the natural map

$$X \to X/\sim \qquad x \mapsto [x]$$

Warning: Each equivalence class may have different names as $[x] = [y]$.

Remark: Conversely, any partition of X arises from a unique equivalence relation.

Now, back to the integers! For $a, b \in \mathbb{Z}$, we have $a \equiv_n b$ if $n|(a-b)$ then $\equiv_n$ is an equivalence relation on $\mathbb{Z}$ (exercise!). In this case, we have n equivalence classes, and so we have

$$[0], [1], ..., [n-1]$$

But the representatives are not unique (so it might be the case that $[0] = [n] = [2n]$)

Definition. We define the set

$$\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/\equiv_n = \{\text{representatives of } \mathbb{Z} \text{ under } \equiv_n\}$$

For now this is a set, but now we will uncover the binary operation that makes this a group. To this end, we equip this set with

$$\bar{a} + \bar{b} = \overline{a+b}$$

and

$$\bar{a}\bar{b} = \overline{ab}$$

One can check that this structure is well defined. More on this in the textbook. Idea is to check if $\bar{a} = \bar{a}'$ and $\bar{b} = \bar{b}'$ then $\overline{a+b} = \overline{a'+b'}$

Example. In $\mathbb{Z}/7\mathbb{Z}$, we have

$$\bar{3} + \bar{5} = \bar{8} = \bar{1}$$

Which is equivalent to $\overline{10} + \overline{(-9)} = \bar{1}$.

We can also try

$$\bar{3} \cdot \bar{5} = \overline{15} = 1$$

which is the same as

$$\overline{3} \cdot \overline{(-2)} = \overline{-6} = \overline{1}$$

Note: $(\mathbb{Z}/n\mathbb{Z}, +)$ is an abelian group (inherits properties from $\mathbb{Z}$).

While $(\mathbb{Z}/n\mathbb{Z}, +)$ is a group, $(\mathbb{Z}/n\mathbb{Z}, \times)$ is not a group due to the lack of inverses, to make this a group, we define the following

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{a} \in \mathbb{Z}/n\mathbb{Z} : \bar{a} \cdot \bar{b} = 1 \text{ for some } \bar{b} \in \mathbb{Z}/n\mathbb{Z}\}$$

What we did here is that we are considering only the elements with an inverse, now this gives us the group structure that we need. In fact, $(\mathbb{Z}/n\mathbb{Z}, \times)$ is an abelian group.

Exercise: Compute the order of the elements in $(\mathbb{Z}/9\mathbb{Z})^\times$.

Lots of Examples and Maps between Groups

Last time, we discussed groups! A set G with a binary operation that is associative, has an identity, and an inverse. Let us talk about Symmetric groups (1.3 in D&F), recall that these groups are denoted by

$$S_n = \text{Permutations of } \{1, ..., n\}$$

We can represent these permutations using cycles. So, in the first cycle we have (1 3 5 7). And similarly the second cycle is a 2-cycle, (2 6) = (6 2), and for the last one it is just 4.

So, we have (1 3 5 7)(2 6) $\in S_7$. Note that these cycles are known as **disjoint cycle decomposition**. More on these later in the course! Now, let us write down the elements of S_3. In this group we have

$$S_3 = \{e, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$$

In principle, we can do this for larger n but it is really tedious since $n!$ grows really fast. Now, let us look at the composition S_5. Consider

$$(1\ 2\ 3) \cdot (1\ 3\ 5) = (1)(2\ 3\ 5)(4)$$

Where we can omit 4 if we want, so the output is (2 3 5). Now let us compute an inverse, consider

$$(1\ 2\ 4\ 3)^{-1} = (1\ 3\ 4\ 2)$$

Note that S_n is not abelian for $n \geq 3$. To prove this, we can do a quick computation with 3 numbers. Say (1 2)(2 3) = (1 2 3) and (2 3)(1 2) = (1 3 2) which are not equal. But, disjoint cycles commute. So what happened is we took a product of cycles that share a common element, but if they don't then they commute!

Reminder: An optional homework problem, if a and b commute, then the same is true for any powers of a and b. This is nice because it becomes nicer to do computation. In general symmetric groups are complicated, they are not commutative, but we use disjoint cycles to make our lives easier.

The next group we will discuss is the dihedral group. Another group that is not abelian!

Definition. The **Dihedral group** is given by

$$D_{2n} = \text{Symmetries of a regular } n\text{-gon in } \mathbb{R}^2$$

Where $n \geq 3$.

We can think of this in terms of the rigid motions of $\mathbb{R}^3$. Suppose we have a regular 5-gon. So we have n rotations about the origin, given by the angle $e^{\frac{2\pi i k}{n}}$. We can also have the reflections, geometrically speaking this might need us to be careful when thinking of this, since the axis goes through the midpoint. In total, we have n reflections, if n is odd, then one axis through each vertex, and when n is even, things are different. We go through two of the vertices, once we go through one we have to go through the other, which gives $n/2$ and then we go through the midpoints to get $n/2$ which adds up to n.

Let r = rotation by $2\pi/n$, so then all rotations are given by r^k for $0 \leq k < n$. In particular, $r^n = 1$, so the order of r is n. Let s denote the reflections, the order of this in the symmetry group is 2. We make a claim:

Claim:

$$D_{2n} = \{1, r, \ldots, r^{n-1},\ s, rs, \ldots, r^{n-1}s\}$$

All distinct by the cancellation law. One way of seeing it is by looking back at linear algebra, as the determinant of $1, ..., r^{n-1}$ is 1, and if we compose $s, ..., r^{n-1}s$ we have determinant -1. So the RHS of the claim has $2n$-distinct elements and the LHS has the same, so we have a complete description of the elements.

Claim: $sr^{-k} = r^k s \quad$ for all $k \in \mathbb{Z}$.

Here we can think in $\mathbb{C} \cong \mathbb{R}^2$. So here our elements are given by

$$r(z) = e^{2\pi i/n} z, \qquad s(z) = \bar{z}$$

Say the center of the polygon at z_0 and one vertex at $z = 1$

In this way, we can get the whole multiplication table, which we will not fully write down, but we can look at some examples

Example. We compute the product $(r^2 s) \cdot (r^3 s)$. Indeed,

$$(r^2 s)(r^3 s) = r^2 (s r^3) s = r^2 (r^{-3} s) s = r^{-1} s^2 = r^{-1}.$$

So $(r^2 s)(r^3 s) = r^{-1} = r^{n-1}$.

Remark: D_{2n} is not abelian, so

$$sr = r^{-1} s \neq rs$$

Important takeaway:

$$sr^k = r^{-k} s \quad \forall k \tag{$*$}$$

Suppose we don't know geometry, how can we define D_{2n}? The group is generated by 2 elements, in fact, every relation between r and s can be derived from $(*)$.

Definition. We write $D_{2n} = \langle r, s | r^n = s^2 = 1,\ sr = r^{-1} s \rangle$

This is called a group presentation, more on this towards the end of group theory! It is hard to determine a group from a given group presentation. Again, we will talk about this more rigorously later! So it might be a bit hard to describe the dihedral group. We will also talk about free groups, which are relevant in Algebraic Topology.

$$\langle x, y | \ \rangle$$

Now, let us move onto matrix groups. Recall that if k is any field, the following are groups

$$(M_{n \times m}(k), +)$$

$$(\mathrm{GL_n(k)}, \times)$$

If we choose certain fields we can get some nice finite groups.

Example. If $k = \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, we get $\mathrm{GL}_n(\mathbb{F}_p)$. Let us compute the order for $n = 2$

$$|\mathrm{GL}_2(\mathbb{F}_p)|$$

We count the number of invertible 2×2 matrices over $\mathbb{F}_p$ by choosing the columns to be linearly independent.

- The first column can be any nonzero vector in $\mathbb{F}_p^2$, there are $p^2 - 1$ choices.

- The second column can be any vector not a scalar multiple of the first (i.e., not in the 1-dimensional subspace it spans), so there are $p^2 - p$ choices.

Hence

$$|\mathrm{GL}_2(\mathbb{F}_p)| = (p^2-1)(p^2-p).$$

Our last example for now is the Quaternion group

Definition.

$$\mathbb{H} = \{x + yi + zj + wk : \ x, y, z, w \in \mathbb{R}\}$$

Note that i, j, k commute with $\mathbb{R}$, moreover $i^2 = k^2 = j^2 = ijk = -1$. Multiplication here is associative but not commutative! .

Example. $ij = k$, since $ij(-k^2) = -ijk \cdot k = k$.

Definition.

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

Note that we haven't shown that anything is well defined just yet, but one can use a multiplication table or (maybe a computer program!). Now that we have seen some examples of groups, we are missing two things: maps between groups and subgroups. We start by discussing isomorphisms and homomorphisms (section 1.6 in Dummit and Foote).

Definition. If $(G, *)$ and $(H, \circ)$ are groups. A **homomorphism** is a function $\varphi : G \to H$ where

$$\varphi(g * h) = \varphi(g) \circ \varphi(h) \ \ \forall g, h \in G$$

Definition. An **isomorphism** is a bijective homomorphism.

Remark: If φ is an isomorphism, then so is the inverse φ^{-1}. Therefore an isomorphism is the same as a homomorphism φ such that there is an inverse $\psi : H \to G$ such that $\varphi \circ \psi = \mathrm{id}_H$ and $\psi \circ \varphi = \mathrm{id}_G$.

We write $G \cong H$, so G and H have the same structure.

Example. If G and H are groups with one element, then $G \cong H$.

Remark: Just like in linear algebra, if we have $G \to H \to K$, if φ and ψ are homomorphisms, then so is $\varphi \circ \psi$, and the same could be said for isomorphisms.

Proposition. If $\varphi : G \to H$ is a homomorphism, then

1. $\varphi(e_G) = e_H$
2. $\varphi(x^{-1}) = \varphi(x)^{-1}$
3. $\varphi(x^n) = \varphi(x)^n$

Proof. We prove (1) only, the rest are left as an exercise. Note that

$$\varphi(e_G) = \varphi(e_G * e_G).$$

Since φ is a homomorphism,

$$\varphi(e_G * e_G) = \varphi(e_G) \circ \varphi(e_G).$$

So

$$\varphi(e_G) = \varphi(e_G) \circ \varphi(e_G).$$

Multiplying both sides on the left (or right) by $\varphi(e_G)^{-1} \in H$ gives

$$e_H = \varphi(e_G)^{-1} \circ \varphi(e_G) = \varphi(e_G)^{-1} \circ \big(\varphi(e_G) \circ \varphi(e_G)\big) = \big(\varphi(e_G)^{-1} \circ \varphi(e_G)\big) \circ \varphi(e_G) = \varphi(e_G).$$

Hence $\varphi(e_G) = e_H$.

□

Example. $(\mathbb{R}, +) \to (\mathbb{R}_+^\times)$ via $x \mapsto e^x$ is an isomorphism

Example. $\mathbb{Z}/6\mathbb{Z} \ncong S_3$, since $\mathbb{Z}/6\mathbb{Z}$ is abelian but S_3 is not.

Example. $S_3 \cong D_6$. We will prove this soon!

Example. $D_8 \ncong Q_8$. We will prove this in HW.

Example. $\mathbb{C}^\times \to \mathrm{GL}_2(\mathbb{R})$ via $x + iy \mapsto \begin{pmatrix} x & y \\ -y & x \end{pmatrix}$ is a homomorphism but not an isomorphism.

Group Actions

Last time, we talked about isomorphisms and homomorphism. If G and H are groups, a homomorphism as a map $\varphi : G \to H$ which preserves the operations. Moreover, an isomorphism is a homomorphism that has an inverse homomorphism. In topology and analysis, these isomorphisms would also have to be continuous (a continuous bijection with a continuous inverse, otherwise known as a homeomorphism). Moreover, we also write $G \cong H$ if there exists an isomorphism $G \to H$. Recall that conjugation is the map $G \to G$, $g \mapsto xgx^{-1}$ for any fixed $x \in G$.

However, the map $G \to G$ given by $g \mapsto g^{-1}$ is not an isomorphism or even a homomorphism but it is if G is abelian. In mathematics, we like to classify things up to isomorphism, so in this course we will classify some groups up to isomorphism.

Definition. If G, H are groups. We can form the **direct product** $G \times H$ with binary operation $(g_1, h_1)(g_2, h_2) = (g_1 * g_2,\ h_1 \circ h_2)$ where $*$ is the operation of G and $\circ$ is the operation of H and the identity is given by $(e_G, e_H) = (1, 1)$.

This allows us to construct more homomorphisms. For example, the projection map $G \times H$ projecting onto either G or H is a homomorphism. Moreover, there is an isomorphism $G \times H \to H \times G$ mapping $(g, h) \mapsto (h, g)$. Now, let us discuss group actions. This rises because when we have a symmetry, we can apply it to our object (e.g a cube), and we can move the object. If we have a permutation, we take one of our object, apply the symmetry and we get a new object.

Definition. A **(Left) group action** of a group G on a set A is a map

$$G \times A \to A, \qquad (g, a) \mapsto g \cdot a (= ga)$$

such that

1. $1 \cdot a = a$ for all $a \in A$
2. $g \cdot (h \cdot a) = (g \cdot_G h) \cdot a$ for all $g, h \in G$ and $a \in A$.

We say that G **acts** on A and write $G \curvearrowright A$

Of course this tells us that there is also the notion of a right action, what changes is that we take

$$(a \cdot h)g = a \cdot hg$$

In this course this won't pose a problem since most of the actions we will work with are left group actions.

Example. $\mathrm{GL}_n(k)$ acts on k^n (column vectors) by matrix multiplication

$$A \cdot v = Av \qquad \text{(Matrix multiplication)}$$

Indeed, $I_n v = v$ and $(AB)v = A(Bv)$

Example. $\mathrm{GL_n}(k)$ acts on k by the determinant

$$A\lambda = \det A\lambda$$

Example. Furthermore, $\mathrm{GL_n}(k)$ acts on ordered (or unordered) pairs of vectors $(k^n \times k^n)$

Example. S_n acts on $\{1, ..., n\}$

$$\sigma \cdot a = \sigma(a)$$

Example. G acts on itself in several ways:

1. $g \cdot a = ga$ for all $g, a \in G$. This is the left regular/translation action.
2. $g \cdot a = ag^{-1}$. This is the right regular action.
3. $g \cdot a = gag^{-1}$. This is the conjugation we defined earlier.
4. Trivial action: $g \cdot a = a$ for all g and a.

And many others.

Another way of thinking of actions is via permutation representations. If $G \curvearrowright A$, then for any $g \in G$. We get a function $\varphi_g : A \to A$ mapping $a \mapsto ga$. Now, the group action means $1 \cdot a = a$ so $\varphi_1 = \mathrm{id}_A$ and $g(h \cdot a) = (gh) \cdot a$ means $\varphi_g \circ \varphi_h = \varphi_{gh}$. This looks like a homomorphism, but we have to know from where to where. We claim that

$$\varphi_g \in S_A.$$

Namely, a bijective function $A \to A$. We just need to show that there is an inverse function. Which in this case we have $\varphi_g \circ \varphi_{g^{-1}} = \varphi_1 = \mathrm{id}_A$, same reasoning can be used to show that $\varphi_{g^{-1}} \circ \varphi_g = \mathrm{id}_A$. So we have that

$$G \to S_A, \qquad g \mapsto \varphi_g$$

is a group homomorphism, and this is the permutation representation of the action.

Proposition. There is a bijection

$$\{\text{Actions of } G \text{ on } A\} \to \{\text{Homomorphisms } G \to S_A\}$$

Proof. Exercise. □

Example. D_{2n} acts on the vertices of the regular n-gon. So, we have n vertices. We get a permutation representation $D_{2n} \to S_n$, which is a homomorphism. When $n = 3$, we get the homomorphism

$$D_6 \to S_3$$

Think of the rotation by $2\pi/3$, we get the 3-cycle (1 2 3) and if we reflect we get (2 3).

Subgroups

Last time we talked about group actions, namely for G a group and A a nonempty set, an action is a map $G \times A \to A$ such that $1 \cdot a = a$ and $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$. Some of the examples we discussed included $S_n \curvearrowright \{1, \dots, n\}$ and $\mathrm{GL}_n(k) \curvearrowright k^n$. Another example that we mentioned was that a group G acts on itself via conjugation, $g \cdot a = gag^{-1}$.

Moreover, we also talked about an equivalent way of defining a group action, namely via the permutation representation: an action $G \curvearrowright A$ is the same as a homomorphism $G \to S_A$, $g \mapsto \varphi_g$, where $\varphi_g(a) = g \cdot a$.

Definition. Let $(G, *)$ be a group. A subset H is a subgroup if $(H, *)$ satisfies the group axioms.

Note that $(H, *)$ here refers to applying the binary operation on elements of H

If H is a subgroup of G, we write $H \leq G$.

Warning: Sometimes Florian will write $H \subset G$ or $H \subseteq G$, but the meaning will be explained whenever he alternates between the notations.

Example.

1. $\{1\}$ is a subgroup of any group.
2. G is a subgroup of itself.

These are the most natural examples of subgroups.

Remark: The subgroup relation is transitive. Namely, if $K \leq H$ and $H \leq G$, then $K \leq G$. (Indeed, K is a group under the operation of H, which is the operation of G.)

Example. More complicated examples include

1. $\{1, r, \dots, r^{n-1}\} \leq D_{2n}$
2. $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ (under addition)
3. $M_n(\mathbb{R}) \leq M_n(\mathbb{C})$ (under addition)
4. $\mathrm{GL}_n(\mathbb{R}) \leq \mathrm{GL}_n(\mathbb{C})$ (under matrix multiplication)

Claim: If $H \leq G$, then $e_H = e_G$ (so $e_G \in H$), and for $h \in H$ the inverse h^{-1} of h in G lies in H and is the inverse of h in H.

Proof. Suppose that e_H is the identity of H. Then

$$e_H e_G = e_H = e_H e_H$$

where the first equality holds since e_G is the identity in G, and the second since e_H is the identity in H. So the cancellation laws in G hold and thus $e_H = e_G$.

For inverses, write h_H^{-1} and h_G^{-1}. We want to show that these are equal. To this end, note that

$$h_H^{-1} h = e_H = e_G = h_G^{-1} h$$

and cancelling h (in G) gives $h_H^{-1} = h_G^{-1}$. Which proves the claim. □

Proposition. The following are equivalent.

1. $H \leq G$
2. $H \neq \varnothing$ and $h_1, h_2 \in H$ implies that $h_1 h_2^{-1} \in H$
3. $H \neq \varnothing$, and H is closed under products and inverses.

Proof. (1 $\implies$ 2) Note that $h_1, h_2 \in H$ implies $h_1 h_2^{-1} \in H$ as H is a group, and $h_H^{-1} = h_G^{-1}$. Moreover, $H \neq \varnothing$ as the identity is in H.

(2 $\implies$ 3) $H \neq \varnothing$ by assumption. So there exists $x \in H$, and taking $h_1 = h_2 = x$ gives $e = xx^{-1} \in H$. Now for $h \in H$ we have $eh^{-1} \in H$, so H is closed under inverses. Moreover, for $h_1, h_2 \in H$ we have $h_2^{-1} \in H$, and so $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$, so H is closed under products.

(3 $\implies$ 1) is an exercise. □

Example. In S_4 we have a subgroup $\{e, (1\ 2)\}$ by parts 2 and 3 of our proposition (as $(1\ 2)^2 = e$). We also have $\{e, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$. Indeed, each non-identity element is its own inverse, and the product of any two distinct non-identity elements is the third; for instance $(1\ 2)(3\ 4) \cdot (1\ 3)(2\ 4) = (1\ 4)(2\ 3)$.

Proposition. If $\varphi : G \to H$ is a group homomorphism, then

1. $\ker \varphi = \{g \in G : \varphi(g) = 1\}$ is a subgroup of G
2. $\operatorname{im} \varphi = \{\varphi(g) : g \in G\}$ is a subgroup of H

Proof. (1) is an exercise. We prove (2). Indeed, this is an application of the subgroup criterion that we just proved. Clearly $\operatorname{im} \varphi$ is nonempty since it contains $\varphi(1) = 1$. Now, for $h_1 = \varphi(g_1)$ and $h_2 = \varphi(g_2)$ in $\operatorname{im} \varphi$, we have

$$h_1 h_2^{-1} = \varphi(g_1)\varphi(g_2)^{-1} = \varphi(g_1)\varphi(g_2^{-1}) = \varphi(g_1 g_2^{-1}) \in \operatorname{im} \varphi.$$

□

Remark: Any subgroup of G is the image of some homomorphism with target G (if $H \leq G$, it is the image of the inclusion homomorphism $H \to G$, $h \mapsto h$). The analogue fails for kernels. For instance, $\{e, (1\ 2)\} \leq S_3$ is not the kernel of any homomorphism: if $\varphi(k) = 1$ then $\varphi(gkg^{-1}) = \varphi(g)\varphi(k)\varphi(g)^{-1} = 1$, so kernels are closed under conjugation, but $(1\ 3)(1\ 2)(1\ 3)^{-1} = (2\ 3) \notin \{e, (1\ 2)\}$.

A useful and natural example of a subgroup is **centre** of G

$$Z(G) = \{g \in G : gx = xg,\ \forall x \in G\}$$

If G is abelian then $Z(G) = G$. Of course, $Z(G) = G$ if and only if G is abelian. To check that $Z(G) \leq G$, observe that $e \in Z(G)$ since $eg = g = ge$ for all g. Now, let $g_1, g_2 \in Z(G)$ and $x \in G$. Since $g_2 x = x g_2$, multiplying on the left and on the right by g_2^{-1} gives $x g_2^{-1} = g_2^{-1} x$. Hence

$$g_1 g_2^{-1} x = g_1 x g_2^{-1} = x g_1 g_2^{-1},$$

so $g_1 g_2^{-1} \in Z(G)$, and $Z(G) \leq G$ by the subgroup criterion.

Example.

1. $Z(Q_8) = \{\pm 1\}$
2. $Z(S_3) = \{1\}$
3. $Z(\mathrm{GL}_n(k)) = \{\lambda I_n : \lambda \in k^\times\}$

We can use group actions to obtain another important example of subgroups. Indeed, if $G \curvearrowright A$, then for any $a \in A$ we define

$$\mathrm{Stab}_G(a) = \{g \in G : g \cdot a = a\} \leq G$$

Proving that this is a subgroup is an exercise.

Example. $G \curvearrowright G$ in two ways. For left translation, $g \cdot a = ga$, we get $\mathrm{Stab}_G(a) = \{g : ga = a\} = \{1\}$. For conjugation, $g \cdot a = gag^{-1}$, we get $\mathrm{Stab}_G(a) = \{g : gag^{-1} = a\} = \{g : ga = ag\}$, the *centralizer* of a.

Example. $G \curvearrowright$ subgroups of G by conjugation. First we have to show that this is a well defined action, if H is a subgroup, then $g \cdot H = \{ghg^{-1} : h \in H\} = gHg^{-1} \leq G$ (this is an exercise). Then the stabilizer is called the normalizer, which we denote by $N_G(H) = \{g \in G : gHg^{-1} = H\}$.

Related: If $H \leq G$, let $C_G(H) = \{g \in G : ghg^{-1} = h, \forall h \in H\}$ which is equivalent to saying $gh = hg$ for all $h \in H$.

Remark: $C_G(H) \supseteq Z(G)$, $C_G(\{1\}) = G$, $C_G(G) = Z(G)$, $C_G(H) \subseteq N_G(H)$. Indeed, an element commuting with all of G commutes with all of H; every g commutes with 1; $g \in C_G(G)$ means $gx = xg$ for all $x \in G$, which is the definition of $Z(G)$; and if $g \in C_G(H)$ then $gHg^{-1} = \{ghg^{-1} : h \in H\} = H$, so $g \in N_G(H)$.

If $G \curvearrowright A$, then the kernel of the action is all $g \in G$ such that $g \cdot a = a$ for all $a \in A$, so this set really is just

$$\bigcap_{a \in A} \mathrm{Stab}_G(a)$$

This is equivalent to the kernel of the permutation representation, namely $\ker(G \to S_A)$, which will be on our next homework. So, in particular, this is why it is called the kernel of the action. One good example is the centre.

Example. $G \curvearrowright G$ by conjugation so the kernel of the action is $Z(G)$. Indeed, g is in the kernel if and only if $gag^{-1} = a$ for all $a \in G$, i.e. $ga = ag$ for all $a \in G$, i.e. $g \in Z(G)$.

Remark: If $H_i \leq G$ for $i \in I$, then $\bigcap_i H_i \leq G$. This is clear from the subgroup criterion: the intersection contains e, and if $h_1, h_2 \in \bigcap_i H_i$ then $h_1 h_2^{-1} \in H_i$ for every i.

Example. Suppose that $G = S_3$ and $H = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \leq G$. Then $(1\ 2) \notin C_G(H)$ and $(1\ 2) \in N_G(H)$. Moreover, we have

$$(1\ 2)\ (1\ 2\ 3)(1\ 2)^{-1} = (1\ 3\ 2)$$

$$(1\ 2)(1\ 3\ 2)\ (1\ 2)^{-1} = (1\ 2\ 3)$$

so conjugation by $(1\ 2)$ swaps the two 3-cycles and fixes e, hence $(1\ 2)H(1\ 2)^{-1} = H$. In fact $C_G(H) = H$ (the transpositions do not commute with $(1\ 2\ 3)$) and $N_G(H) = S_3$ (as S_3 is generated by $(1\ 2)$ and $(1\ 2\ 3)$).

Example. Let $G = D_{2n}$ with $n \geq 3$. Then $C_G(r) = \langle r \rangle = \{1, r, \ldots, r^{n-1}\}$. Indeed, rotations commute with r. For a reflection $r^k s$ we have $(r^k s)r = r^k r^{-1} s = r^{k-1} s$ while $r(r^k s) = r^{k+1} s$, and these agree only if $r^2 = 1$, which is impossible since r has order $n \geq 3$.

Definition. A group G is cyclic if G can be generated by a single element. That is, to say $G = \{x \in G : x^n \ n \in \mathbb{Z}\}$

Example. Some examples include:

1. $\{1, r, ..., r^{n-1}\} \leq D_{2n}$
2. $\mathbb{Z}$, this is generated by $x = \pm 1$
3. $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 1$, $x = \bar{1}$ (or even $x = i, \ \gcd(i, n) = 1$)

Example. Some non-examples include:

1. $(\mathbb{Q}, +)$
2. Any uncountable group (for instance $(\mathbb{R}, +)$, $\mathbb{C}^\times$, $S^1 \leq \mathbb{C}^\times$).

Cyclic Groups

Last time, we talked about subgroups, the subgroup criterion, so if we want to check if something is a subgroup, we can use a criterion similar to linear algebra. We also looked at some examples (most notably, the image and kernel of a homomorphism, centralizers and stabilizers). Furthermore, we also defined what a cyclic group is. Namely a group that is generated by 1 element. Examples include $\mathbb{Z}$, $\mathbb{Z}/n\mathbb{Z}$, the group of rotations $\leq D_{2n}$. In general, for $g \in G$ the set of all powers of g, $\langle g \rangle = \{g^k \in G : k \in \mathbb{Z}\}$, is a cyclic subgroup of G (it is the image of the homomorphism $\mathbb{Z} \to G$, $k \mapsto g^k$). Non examples include uncountable groups and $(\mathbb{Q}, +)$.
Our goal now is to classify all cyclic groups up to isomorphism.

Lemma. If G is cyclic, with generator x and $|x| = n \geq 1$. Then

$$G = \{1, x, x^2, \ldots, x^{n-1}\},\ |G| = n$$

Proof. By assumption, $G = \{x^k : k \in \mathbb{Z}\}$. Now, we claim that $x^i = x^j$ if and only if $i \equiv j \pmod{n}$.

($\Rightarrow$) Assume $i \geq j$ (otherwise swap the roles) and $x^i = x^j$. Multiplying by x^{-j} we get $x^{i-j} = 1$. Write $i - j = qn + r$ where $0 \leq r < n$, this is precisely the division with remainder, then we have that

$$1 = x^{i-j} = (x^n)^q x^r = x^r.$$

Since n is the smallest positive integer with $x^n = 1$ and $0 \leq r < n$, we must have $r = 0$. So $n \mid (i - j)$, that is, $i \equiv j \pmod{n}$.

($\Leftarrow$) If $i - j = qn$, then $x^{i-j} = (x^n)^q = 1$, so $x^i = x^j$.

Consequently, every x^k equals x^r where r is the remainder of k modulo n, so $G = \{1, x, \ldots, x^{n-1}\}$. Moreover, these elements are pairwise distinct: if $x^i = x^j$ with $0 \leq i, j \leq n - 1$ then $i \equiv j \pmod{n}$ forces $i = j$. Hence $|G| = n$. □

Remark: If $|x| = \infty$, then $x^i = x^j$ implies $x^{i-j} = 1$ and hence $i = j$, so all powers of x are distinct and $\langle x \rangle$ is infinite. In particular, if $G = \langle x \rangle$ then G is finite if and only if $|x| < \infty$, and in that case $|G| = |x|$.

Proposition.

1. If G is cyclic, and $|G| = n$, then $G \cong \mathbb{Z}/n\mathbb{Z}$
2. If G is cyclic and $|G| = \infty$, then $G \cong \mathbb{Z}$

Proof. In the first case, write $G = \langle x \rangle$. By the remark above, $|x| = |G| = n$. Define a map

$$\varphi : \mathbb{Z}/n\mathbb{Z} \to G, \qquad \bar{i} \mapsto x^i$$

Since we are mapping from an equivalence class, we have to show that φ is well defined. To this end note that $\bar{i} = \bar{j}$ implies $x^i = x^j$ thanks to our previous claim. Now, we want to show that this map is a homomorphism:

$$\varphi(\bar{i} + \bar{j}) = \varphi(\overline{i + j}) = x^{i+j} = x^i x^j = \varphi(\bar{i})\varphi(\bar{j}).$$

It is surjective since $G = \{x^k : k \in \mathbb{Z}\}$, and it is injective since $x^i = x^j$ implies $i \equiv j \pmod{n}$, i.e. $\bar{i} = \bar{j}$, by the previous claim. Hence φ is an isomorphism.

To prove the second case, $G = \langle x \rangle$, $|x| = \infty$ by the remark after the lemma. Define $\mathbb{Z} \to G$ via $i \mapsto x^i$. The rest of the proof is left as an exercise. □

Proposition. Let G be a group, $x \in G$ and $i \in \mathbb{Z}_{\geq 0}$.

1. If $|x| = \infty$ and $i \geq 1$, then $|x^i| = \infty$
2. If $|x| = n$, then $|x^i| = \dfrac{n}{\gcd(i,n)}$

Proof. (1) is an exercise. We prove (2). Indeed, note that $(x^i)^j = 1$ is equivalent to $x^{ij} = 1$, and this is equivalent to $n|ij$ (by the lemma). Now, we want to think about the gcd, let $d = \gcd(i,n)$, so we write everything as a multiple of the gcd, so

$$n = dn' \qquad i = di'$$

where $\gcd(i',n') = 1$. Then $n|ij$ is equivalent to $dn'|di'j$, which is equivalent to $n'|i'j$. Since $\gcd(i',n') = 1$, this is equivalent to $n'|j$. Hence the smallest positive j with $(x^i)^j = 1$ is $j = n' = n/d$, that is, $|x^i| = n/\gcd(i,n)$. □

Theorem. Suppose that $G = \langle x \rangle$ is cyclic. Then

1. Every subgroup of G is cyclic
2. If G is infinite, then the subgroups are $\langle x^m \rangle$ for $m \geq 0$.
3. If $|G| = n$ is finite, then the subgroups are $\langle x^d \rangle$ for $d \geq 1$ with $d|n$, and $\langle x^d \rangle$ has order n/d.

Example. The subgroups of $\mathbb{Z}$ are $\langle m \rangle = m\mathbb{Z}$, $m \geq 0$, and the subgroups of $\mathbb{Z}/n\mathbb{Z}$ are $\langle \bar{d} \rangle = d\mathbb{Z}/n\mathbb{Z}$ where $d \geq 1$, $d|n$

Proof of the theorem. The case where $|G| = \infty$ is an exercise.

Assume $|G| = n$, namely $|x| = n$. Let $H \leq G$. Since $x^n = 1 \in H$, we can let $d \geq 1$ be minimal such that $x^d \in H$, which implies that $\langle x^d \rangle \subseteq H$. We claim that there is an equality, and d divides n. Indeed, if $x^i \in H$, then we can write $i = dq + r$, and $0 \leq r < d$, then we know

$$x^r = x^i (x^d)^{-q} \in H.$$

By minimality of d we must have $r = 0$, so $d \mid i$ and $x^i = (x^d)^q \in \langle x^d \rangle$. Hence $H = \langle x^d \rangle$, and in particular H is cyclic. Applying this to $i = n$ (as $x^n = 1 \in H$) gives $d \mid n$. Finally, by the previous proposition, $|\langle x^d \rangle| = |x^d| = n/\gcd(d,n) = n/d$. Conversely, for every $d \geq 1$ with $d|n$, $\langle x^d \rangle$ is a subgroup of order n/d, and different divisors give different orders, hence different subgroups. □

Quotients...Hello darkness my old friend

Last time, we talked about cyclic groups, which are the groups that are generated by 1 element. Moreover, we classified cyclic groups up to isomorphism.

Exercise: If $|G|$ is finite then $\langle x^d \rangle \subset \langle x^e \rangle$ for $de|n$ if and only if $e|d$.

Today, we will talk a little more about subgroups generated by a subset and then we will move to quotient groups. If $A \subset G$ is a subset, we want the smallest subgroup containing A. So the most elegant way to do it is to look at the collection of all subgroups containing A. The idea is to take the intersection, more precisely

$$\langle A \rangle = \bigcap_{H \leq G H \supset A} H$$

Lemma. If $H_i \leq G$ for $i \in I$. Then

$$\bigcap_{i \in I} H_i$$

Proof. Exercise. Use the subgroup criterion. □

Then $\langle A \rangle \subset H$ for all $H \leq G$ by definition, so $\langle A \rangle$ is the smallest subgroup generated by A.

Lemma. $\langle A \rangle = \{a_1^{\varepsilon_1} a_2^{\varepsilon_2} \cdots a_n^{\varepsilon_n} : a_i \in A, \varepsilon_i \in \{\pm 1\},\ n \geq 0\}$

Proof. " $\supset$ " is clear, since $(a_i \in H \implies a_1^{\varepsilon_1} a_2^{\varepsilon_2} \cdots a_n^{\varepsilon_n} \in H)$. Conversely, by the subgroup criterion the right hand side is a subgroup, so $\langle A \rangle \subset$ RHS by definition of $\langle A \rangle$ being the smallest subgroup. □

Example. $\langle \{a\} \rangle = \{a^n : n \in \mathbb{Z}\}$.

Example. $\langle \{a, b\} \rangle = \{a^{n_1} b^{m_1} a^{n_2} b^{m_2} \cdots : n_i, m_i \in \mathbb{Z}\}$. Note that the product here is finite.

Let us now talk about the lattice of subgroups. Suppose now that G is finite (not too big), consider the group $\mathbb{Z}/30\mathbb{Z}$, the subgroups we get are $\langle \bar{1} \rangle, \langle \bar{2} \rangle, \langle \bar{3} \rangle, \langle \bar{5} \rangle, \langle \bar{6} \rangle, \langle \bar{10} \rangle, \langle \bar{15} \rangle, \langle \bar{30} \rangle$.

Now we shift gears towards quotient groups. The nice thing is we already constructed such a group earlier in the course. So the idea is that we will generalize the construction $\mathbb{Z}$ to $\mathbb{Z}/N\mathbb{Z}$.

Given G we want equivalence $\sim$ such that $G/\sim$ (set of equivalence classes of G) is a group under

$$\overline{g} \cdot \overline{h} = \overline{gh} \qquad (\bigstar)$$

So we want to classify all equivalence relations such that this holds. If this is a group, let $N = \bar{1} \subset G$, note that $\bar{1}$ here denotes the equivalence class of the identity $\{g \in G : g \sim 1\}$. We claim that N is a subgroup. Indeed, we have a subgroup criterion so we can put that to work, clearly this is nonempty as $1 \in N$. So we show that if $n_1 n_2 \in n$ then $\bar{n}_1 = \bar{1} = \bar{n}_2$ so $\bar{n}_1 \bar{n}_2 = \overline{n_1 n_2} = \bar{1}$ as $\bar{1}\bar{1} = \overline{1 \cdot 1} = \bar{1}$. Thanks to $(\bigstar)$. The inverse is left as an exercise.

Remark: or, we get $\pi : G \to G/\sim$ a group homomorphism via $g \mapsto \bar{g}$, and $N = \ker \pi$.

Claim: Given $g \in G$, $n \in N$, we have $gng^{-1} \in N$.

Proof. By (✯), $\overline{gng^{-1}} = \bar{g} \cdot \bar{n} \cdot \bar{g}^{-1} = \bar{g}\bar{g}^{-1} = 1$, again thanks to (✯). □

Remark: If $g \in G$, $n \in \ker \varphi$, then $gng^{-1} \in \ker \varphi$

Claim: For any $g \in G$, $\bar{g}$, we have $\bar{g} = \{gn : n \in N\} =: gN$.

Proof. To see that, we want to know when an element is contained in the equivalence class of g, namely $g \sim h$. How can we relate that to when it is equal to the class of the identity. So $g \sim h \iff \bar{g} = \bar{h}$. Which is then $\overline{g^{-1}h} = \bar{1}$. Using (✯), we get $g^{-1}\bar{h}$, hence $(\bar{g})^{-1}\bar{h}$. Equivalently $g^{-1}h \in N$, and so $h \in gN$. □

Remark: gN is referred to as a coset.

Definition. A subgroup $N \leq G$ is said to be **normal** if $gng^{-1} \in G$ for any $g \in G$, $n \in N$. Moreover, we denote by $N \lhd G$

Example.

1. Any subgroup N of G is normal if G is abelian.
2. $G \lhd G$
3. $\{1\} \lhd G$
4. $\ker \varphi \lhd G$ if $\varphi : G \to H$ is a group homomorphism. Indeed if $\varphi(n) = 1$, then $\varphi(gng^{-1}) = \varphi(g)\varphi(n)\varphi(g^{-1}) = 1$

Example. Some non examples include:

1. $< (1\ 2) > \not\lhd S_3$. Indeed, we just want to find a way to conjugate $(1\ 2)$ such that we do not get $(1\ 2)$. So we can compute $(2\ 3)(1\ 2)(1\ 2)^{-1} = (1\ 3)$.

Lemma. The following are equivalent:

1. $N \lhd G$
2. $gNg^{-1} = N$ for all $g \in G$
3. $gNg^{-1} \subset N$ for all $g \in G$
4. $gN = Ng$ for all $g \in G$.

Proof.
1. $(1 \iff 3)$ is trivial
2. $(2 \iff 4)$, we multiply by g
3. $(2 \to 3)$ is trivial.
4. $(3 \implies 2)$ apply (3) for g^{-1}.

□

Remark/Exercise: $H \lhd G$ if and only if $N_G(H) = G$.

Now, we can construct quotient groups. Suppose that $N \lhd G$ is a normal subgroup. Define a relation $\sim$ on G by (whatever we did previously), so $g \sim h$ if and only if $g^{-1}h \in N$. So now we have to check that this is an equivalence

relation (this is easy but we will skip it, do note that we will not need normality so just check that $\bar{g} = gN$). Now that we have an equivalence relation, this forms a partition of G of disjoint cosets.

Definition. The **quotient group** G/N is given by

$$\begin{aligned} G/N &= \{\bar{g} : g \in G\} \\ &= (\{gN : g \in G\}, \bigstar) \\ &= gN \cdot hN = ghN \end{aligned}$$

Theorem. G/N is a group with operation $(\bigstar)$.

Proof. First, we prove that $(\bigstar)$ is well defined. To this end, suppose that $gN = xN$ and $hN = yN$, so we want

$$ghN = xyN$$

so

$$ghN = gyN = g(Ny) = xNy = xyN$$

thanks to normality and the lemma where $gN = Ng$. Therefore, the operation is well defined.

So now we have a well defined operation on the set of equivalence classes. It remains to show that G/N satisfies the group axioms. Indeed, it suffices to note that G/N inherits the group structure from G. More precisely

1. (identity): $1 \cdot N = N$
2. (inverse): $(gN)^{-1} = g^{-1}N$
3. (associativity): $[(gN)(hN)](kN) = (gh)kN$

Thus G/N is a group. □

Remark: By $(\bigstar)$, we have a surjective group homomorphism

$$\pi : G \to G/N, \qquad g \mapsto gN = \bar{g}$$

Moreover, $\ker \pi = N$. In particular, any normal subgroup of G is the kernel of some homomorphism $G \to H$.

Example. If $G = \mathbb{Z}$, $N = n\mathbb{Z}(\triangleleft G)$, then $\mathbb{Z}/n\mathbb{Z}$ is the same as before!

Example. If G is abelian, then any subgroup N is normal. Moreover, G/N is abelian.

Example. $G/G \cong \{\bar{1}\}$, similarly, if we quotient G out by the trivial normal subgroup, we get $G/\{1\} \cong G$.

Example. $\langle(1\ 2\ 3)\rangle \triangleleft S_3$. note that $<1\ 2\ 3>$ has order 3 while S_3 has order 6. So we have $S_3/\langle 1\ 2\ 3\rangle$is cyclic with order 2.

Now, if $H \leq G$ is any subgroup, we can define the left and right cosets. So, the left coset gH and the right coset Hg, usually these are not the same unless H is normal. These are equivalence classes for $g_1 \sim g_2 \iff g_1^{-1}g_2H = H$ in the former while the equivalence classes in the latter are given by $g_1 \sim g_2 \iff g_1g_2^{-1}$

Lagrange's Theorem and the First Isomorphism Theorem

Unfortunately I was not able to make it to class (I hate MiWay and the TTC). This section should provide a summary of what was covered on Friday. That aside, on Thursday we talked about normal subgroups and quotient groups, namely

$$\begin{aligned} G/N &= \{\bar{g} : g \in G\} \\ &= (\{gN : g \in G\}, \bigstar) \\ &= gN \cdot hN = ghN \end{aligned}$$

Where $\bigstar$ is the operation defined on Thursday and $N \lhd G$. Moreover, there is an isomorphism $G \to G/N$ via $g \mapsto [g]$. More generally, we have for $H \leq G$, we have the left cosets gH and right cosets Hg, which are both partitions of G.

Lemma.

$$\#\{\text{Left cosets}\} = \#\{\text{Right cosets}\}$$

Proof. Use the map

$$gH \mapsto Hg^{-1}$$

This is invertible with inverse

$$Hg \mapsto g^{-1}H$$

So this is clearly a bijection. □

Definition. The index of $H \leq G$ is said to be the number of left (or) right cosets, we denote this by $(G : H)$

Lemma. If G is finite, then $|G| = (G : H) = |H|$.

Proof. Note that we have a bijection from the left cosets to the right cosets. Namely

$$H \to xH \qquad h \mapsto xh \qquad \text{(with an inverse element } xH \mapsto x^{-1}H\text{)}$$

so $|xH| = |H|$. Moreover, $\#\{\text{cosets}\}|H| = |G|$. Implying that

$$(G : H)|H| = G$$

As required. □

Remark: In particular, $|H|$ divides $|G|$, namely Lagrange's theorem.

Corollary. If $a \in G$ and G is finite, then $|a|$ divides $|G|$.

Proof. Since $|\langle a \rangle| = |a|$, and $\langle a \rangle \leq G$. The result follows by Lagrange's theorem. □

Corollary. If $|G| = p$, then G is cyclic.

We prove another lemma

Lemma. If $H \lhd G$, then $HK = KH$, so $HK \leq G$.

Proof. It suffices to see that

$$HK = \bigcup_{k \in K} HK = \bigcup_{k \in K} KH = KH.$$

As required. □

Lemma. If $H \leq G$ has index 2, then $H \lhd G$.

Proof. Note that since cosets partition G, it follows that we have 2 left and right cosets $H, G \setminus H$. In other words, the number of left and right cosets is the same, as a result $H \lhd G$. □

Now, we revisit an old friend from linear algebra. Recall that if $\varphi : V \to W$ is linear, then

$$\dim \ker \varphi + \dim \operatorname{im} \varphi = \dim V$$

Furthermore, we saw something even more general. Namely

$$V / \ker \varphi \cong \operatorname{im} \varphi$$

This is the First Isomorphism Theorem for vector spaces. Of course, this holds for groups as well

Theorem. **(First Isomorphism Theorem)** If $\varphi : G \to H$ is a group homomorphism, then

$$G / \ker \varphi \cong \operatorname{im} \varphi$$

with $\bar{g} \mapsto \varphi(g)$.

Proof. Let $\Psi : G / \ker \varphi \to \operatorname{im} \varphi$, $\Psi[g] = \varphi(g)$. First, we prove that this is well defined. Indeed, let $g_1, g_2 \in [g]$. Then

$$\Psi([g_1]) = \varphi(g_1)$$

and

$$\Psi([g_2]) = \varphi(g_2)$$

Note that $g_1 \sim g_2$ implies that $g_1 \ker \varphi = g_2 \ker \varphi$. Implying that

$$\varphi(g_1 \ker \varphi) = \varphi(g_2 \ker \varphi)$$

so

$$\varphi(g_1) \ker \varphi = \varphi(g_2) \ker \varphi.$$

Thus

$$\varphi(g_1) = \varphi(g_2)$$

Therefore it is well defined. Now, we claim that this is an isomorphism. To this end, we first show that this is a group homomorphism. Indeed, note that $\Psi([1]) = \varphi(1) = 1$, and

$$\begin{aligned} \Psi([g][h]) &= \Psi([gh]) \\ &= \varphi(g) \cdot \varphi(h) \\ &= \Psi([g])\Psi([h]) \end{aligned}$$

Hence Ψ is a group homomorphism. Now, it remains to prove that Ψ is bijective. Indeed, clearly this map is surjective (by definition of a quotient map). Injectivity follows from $\varphi(g_1) = \varphi(g_2)$ implies that $\varphi(g_1 g_2 - 1) = e$. Thus Ψ is an isomorphism. □

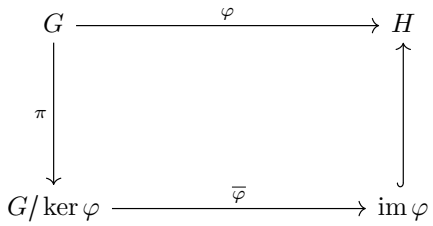
More Isomorphism Theorems, Alternating Groups, and Simple Groups

Last week, we talked about normal subgroups and quotient groups. Moreover, we started talking about isomorphism theorems, in particular, we proved the first isomorphism theorem, which states that if $\varphi : G \to H$ is a group homomorphism, then

$$G/\ker\varphi \cong \operatorname{im}\varphi, \qquad \bar{g} \mapsto \varphi(g).$$

Today, we will continue talking about them.

Remark: The first isomorphism theorem shows that the following diagram commutes.



Corollary. A homomorphism φ is injective if and only if $\ker\varphi = \{1\}$.

Proof. ($\Rightarrow$) If φ is injective and $\varphi(g) = 1 = \varphi(1)$, then $g = 1$, so $\ker\varphi = \{1\}$.

($\Leftarrow$) Suppose that $\ker\varphi = \{1\}$. Then the quotient map $\pi : G \to G/\ker\varphi = G/\{1\}$ is an isomorphism. By the remark above, φ is a composition of three maps

$$G \xrightarrow{\pi} G/\ker\varphi \xrightarrow{\overline{\varphi}} \operatorname{im}\varphi \hookrightarrow H,$$

each of which is injective (π and $\overline{\varphi}$ are isomorphisms, and the last map is an inclusion). Hence φ is injective.

Alternatively, if $\varphi(g_1) = \varphi(g_2)$, then $\varphi(g_1 g_2^{-1}) = \varphi(g_1)\varphi(g_2)^{-1} = 1$. Since the kernel is trivial, $g_1 g_2^{-1} = 1$, so $g_1 = g_2$. □

Example. Consider $\mathrm{GL}_n(k)/\mathrm{SL}_n(k) \cong k^\times$ (here k is a field). We use the map $\varphi = \det : \mathrm{GL}_n(k) \to k^\times$, which is a homomorphism since $\det(AB) = \det(A)\det(B)$. Note that $\mathrm{SL}_n(k) = \ker\det$, and $\operatorname{im}\varphi = k^\times$ since $\det\operatorname{diag}(\lambda, 1, \ldots, 1) = \lambda$ for every $\lambda \in k^\times$. So the claim follows from the first isomorphism theorem.

Theorem. **(Second Isomorphism Theorem)** If $H \le G$ and $N \lhd G$, then $HN/N \cong H/(H \cap N)$.

Remark: In particular, $HN \le G$, $N \lhd HN$, and $H \cap N \lhd H$.

Remark: All we need is that $N \le G$ and $H \le N_G(N)$.

Proof. Since $N \lhd G$, we have $HN \le G$ and $N \lhd HN$. We construct a homomorphism

$$\varphi : H \to HN/N, \qquad h \mapsto \bar{h} = hN.$$

Then φ is a homomorphism since $\overline{h_1 h_2} = \bar{h}_1 \bar{h}_2$. It remains to compute the kernel and image of this homomorphism.

- $\ker\varphi = \{h \in H : hN = N\} = \{h \in H : h \in N\} = H \cap N$.
- As for $\operatorname{im}\varphi$, we claim that φ is surjective. Indeed, take $\overline{hn} \in HN/N$ with $h \in H$ and $n \in N$. Then $hnN = hN$ since $n \in N$, so $\overline{hn} = \bar{h} = \varphi(h)$.

The first isomorphism theorem (which also shows that $H \cap N \lhd H$, being a kernel) then gives $H/(H \cap N) \cong HN/N$. □

Remark: This is the least useful isomorphism theorem.

Theorem. **(Third Isomorphism Theorem)** If $N \leq K \leq G$ such that $N \lhd G$ and $K \lhd G$, then

$$(G/N)/(K/N) \cong G/K.$$

Remark: In particular, $K/N \lhd G/N$.

Proof. Again, we will use the first isomorphism theorem. Consider the map

$$\varphi : G/N \to G/K, \qquad \bar{g} = gN \mapsto gK.$$

We claim that this map is well defined. To this end, note that $gN = hN$ implies $h^{-1}g \in N \subseteq K$, hence $gK = hK$. Proving that φ is a homomorphism is an exercise (Hint: This is immediate!).

Now, note that $\ker\varphi = \{gN : gK = K\} = \{gN : g \in K\} = K/N$, implying that $K/N \lhd G/N$. Moreover, φ is surjective, since every coset gK equals $\varphi(gN)$, so $\operatorname{im}\varphi = G/K$.

So, by the first isomorphism theorem, we deduce that $(G/N)/(K/N) \cong G/K$. □

Exercise: Prove this theorem by using a map from G to the double quotient.

Theorem. **(Fourth Isomorphism = Correspondence Theorem)** Let $N \lhd G$ and let $\pi : G \to G/N$ be the quotient map. Then we have a bijection

$$\{\text{subgroups of } G/N\} \longrightarrow \{\text{subgroups of } G \text{ that contain } N\}, \qquad \bar{H} \mapsto H = \pi^{-1}(\bar{H}),$$

with inverse $H \mapsto \bar{H} = H/N$. For $\bar{H}_1, \bar{H}_2 \leq G/N$ corresponding to $H_1, H_2 \leq G$, this bijection satisfies:

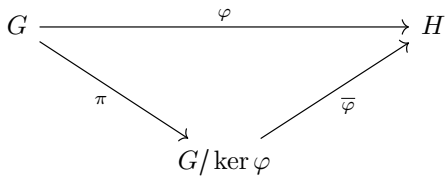
1. Order preserving: $\bar{H}_1 \leq \bar{H}_2$ if and only if $H_1 \leq H_2$.
2. If $\bar{H}_1 \leq \bar{H}_2$, then $\bar{H}_1 \lhd \bar{H}_2$ if and only if $H_1 \lhd H_2$. Moreover, in this case $\bar{H}_2/\bar{H}_1 \cong H_2/H_1$.
3. If $\bar{H}_1 \leq \bar{H}_2$, then $(\bar{H}_2 : \bar{H}_1) = (H_2 : H_1)$.

Remark: In particular, taking $\bar{H}_2 = G/N$ in (2) shows that $\bar{H} \lhd G/N$ if and only if $H \lhd G$, which is the most important case. Furthermore, (1) shows that maximal and minimal subgroups also correspond. More generally, if $\varphi : G \to K$ is a homomorphism and $M \lhd K$, then $\varphi^{-1}(M) \lhd G$ (Exercise!).

Proof. Note that we have the quotient map $\pi : G \to G/N$. The two maps $\bar{H} \mapsto \pi^{-1}(\bar{H})$ and $H \mapsto \pi(H) = H/N$ are inverse to each other. The rest of the details are left as an exercise. □

Remark: For anyone who wants to go into algebra, there is the universal property of the quotient, which says that for $N \lhd G$, if $\varphi : G \to H$ is a homomorphism such that $\varphi(N) = \{1\}$, then there is a unique homomorphism $\bar{\varphi} : G/N \to H$ such that $\bar{\varphi}(gN) = \varphi(g)$ for all $g \in G$. The proof is left as an exercise. In fact, we can use this to make our life easier in the proof of the third isomorphism theorem.

Remark: It is sometimes useful to think categorically, since all we need to look at is groups and maps between them, rather than looking at individual elements. For instance, the diagram below commutes.



We will now talk about alternating groups. In some sense, we have seen the concept before in linear algebra, when discussing determinants, namely in changing signs. We start from the sign of a permutation. Consider a group action $S_n \curvearrowright P_n(\mathbb{C})$, where $P_n(\mathbb{C})$ denotes the polynomials in $X_1, \dots, X_n$ with complex coefficients. Concretely, if $\sigma \in S_n$, let

$$\sigma(p(X_1, \dots, X_n)) = p(X_{\sigma(1)}, \dots, X_{\sigma(n)}).$$

This is a group action, the identity acts trivially, and on the variables we have

$$\sigma(\tau X_i) = \sigma X_{\tau(i)} = X_{\sigma(\tau(i))} = (\sigma\tau)X_i,$$

which extends to all polynomials since substitution respects sums and products.

The trick is to take

$$\triangle = \prod_{i<j}(X_i - X_j).$$

For example, for $n = 3$ we have

$$\triangle = (X_1 - X_2)(X_1 - X_3)(X_2 - X_3).$$

Then for $\sigma \in S_n$, we have

$$\sigma(\triangle) = \prod_{i<j}(X_{\sigma(i)} - X_{\sigma(j)}) = \pm\triangle,$$

since each factor $X_{\sigma(i)} - X_{\sigma(j)}$ is, up to a sign, one of the factors of $\triangle$, and σ permutes the pairs $\{i, j\}$. We can count what the total sign change is, but let us first state what the sign of a permutation is.

Definition. The **sign** of a permutation is the map $\varepsilon : S_n \to \{\pm 1\}$ defined by $\sigma(\triangle) = \varepsilon(\sigma)\triangle$, so that

$$\varepsilon(\sigma) = (-1)^{|\{(i,j):\ i<j,\ \sigma(i)>\sigma(j)\}|}.$$

Indeed, the factor $X_{\sigma(i)} - X_{\sigma(j)}$ is the negative of a factor of $\triangle$ exactly when $\sigma(i) > \sigma(j)$.

Lemma. ε is a homomorphism.

Proof. Since this is a group action, $\sigma(\tau(\triangle)) = (\sigma\tau)(\triangle)$. Now $\tau(\triangle) = \varepsilon(\tau)\triangle$, so

$$(\sigma\tau)(\triangle) = \sigma(\varepsilon(\tau)\triangle) = \varepsilon(\tau)\,\sigma(\triangle) = \varepsilon(\tau)\varepsilon(\sigma)\,\triangle\,.$$

On the other hand, $(\sigma\tau)(\triangle) = \varepsilon(\sigma\tau)\triangle$. Since $\triangle \neq 0$, we conclude that

$$\varepsilon(\sigma\tau) = \varepsilon(\sigma)\cdot\varepsilon(\tau).$$

□

Example. If $\sigma = (r\ s)$ (for $r < s$), then the pairs $i < j$ with $\sigma(i) > \sigma(j)$ are (r, s), together with (r, j) and (j, s) for each $r < j < s$. That is $2(s - r - 1) + 1$ pairs, an odd number, so

$$\varepsilon(\sigma) = -1.$$

Example. If $\sigma = (i_1\ i_2\ \cdots\ i_k)$ is a k-cycle, then $\sigma = (i_1\ i_2)(i_2\ i_3)\cdots(i_{k-2}\ i_{k-1})(i_{k-1}\ i_k)$, a product of $k - 1$ transpositions. Since ε is a homomorphism, the previous example implies that $\varepsilon(\sigma) = (-1)^{k-1}$.

Since every permutation is a product of cycles, hence of transpositions, these examples tell us that σ is an **even** permutation if and only if $\varepsilon(\sigma) = 1$, which is also equivalent to saying that σ is a product of an even number of transpositions.

Definition. We define the **alternating group** as

$$A_n = \ker \varepsilon \lhd S_n, \qquad \text{where } \varepsilon : S_n \to \{\pm 1\}.$$

By the first isomorphism theorem, we have an isomorphism

$$S_n/A_n \cong \operatorname{im} \varepsilon = \begin{cases} \{\pm 1\} & \text{if } n \geq 2, \\ \{1\} & \text{if } n = 1, \end{cases}$$

where for $n \geq 2$ the map ε is surjective since $\varepsilon((1\ 2)) = -1$. So $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$ if $n \geq 2$. Now, we move on to composition series, but first we define what a simple group is.

Definition. We say that G is **simple** if $G \neq \{1\}$ and its only normal subgroups are $\{1\}$ and G.

Example. Examples of simple groups:

1. $\mathbb{Z}/p\mathbb{Z}$ for all primes p, by Lagrange's theorem. Basically, any group of prime order is simple.
2. A_n for any $n \geq 5$ (soon!)

Example. Non examples include:

1. $\mathbb{Z}$ (since $2\mathbb{Z} \lhd \mathbb{Z}$ is a proper nontrivial normal subgroup)
2. S_3 (since $A_3 \lhd S_3$ is a proper nontrivial normal subgroup)

Composition Series, Orbits, and the Orbit-Stabilizer Theorem

Last time, we proved the rest of the isomorphism theorems for groups, and the correspondence theorem, which states that for $N \triangleleft G$ and $\pi : G \to G/N$ the quotient map, we have a bijection

$$\{\text{subgroups of } G/N\} \longrightarrow \{\text{subgroups of } G \text{ that contain } N\}, \qquad \bar{H} \mapsto H = \pi^{-1}(\bar{H}),$$

with inverse $H \mapsto \bar{H} = H/N$. For $\bar{H}_1, \bar{H}_2 \leq G/N$ corresponding to $H_1, H_2 \leq G$, this bijection satisfies:

1. Order preserving: $\bar{H}_1 \leq \bar{H}_2$ if and only if $H_1 \leq H_2$.
2. If $\bar{H}_1 \leq \bar{H}_2$, then $\bar{H}_1 \triangleleft \bar{H}_2$ if and only if $H_1 \triangleleft H_2$. Moreover, in this case $\bar{H}_2/\bar{H}_1 \cong H_2/H_1$.
3. If $\bar{H}_1 \leq \bar{H}_2$, then $(\bar{H}_2 : \bar{H}_1) = (H_2 : H_1)$.

Warning: For $H \leq G$ (not necessarily containing N), we have

$$\begin{aligned} \pi(H) &= \{hN : h \in H\} \\ &= \{hnN : h \in H,\ n \in N\} \\ &= HN/N, \end{aligned}$$

which equals H/N only when $N \leq H$ (in which case $HN = H$).

Furthermore, we also defined the alternating group to be the kernel of the sign map, namely $A_n = \ker(\varepsilon : S_n \to \{\pm 1\})$, and we also defined simple groups, which are the nontrivial groups whose only normal subgroups are the group itself and $\{1\}$. Some important examples include

Example.

1. A_n is simple for all $n \geq 5$.
2. $\mathrm{SL}_n(k)/Z(\mathrm{SL}_n(k))$ is simple for all $n \geq 3$, or $n = 2$ and $|k| \geq 4$ (here k is a field).

and various other groups.

Remark: These groups are really hard to classify! Look up the classification of finite simple groups.

Definition. A **composition series** of G is a sequence

$$\{1\} = N_0 \triangleleft N_1 \triangleleft \cdots \triangleleft N_k = G$$

such that N_i/N_{i-1} is simple for all $i = 1, \ldots, k$. The quotients N_i/N_{i-1} are called the **composition factors**.

Remark: If G is finite, then a composition series exists, by induction on $|G|$. If $G = \{1\}$, take $k = 0$. Otherwise, take a normal subgroup $N \triangleleft G$ with $N \neq G$ and $|N|$ maximal. Then G/N is simple: if G/N had a normal subgroup $\bar{M} \neq \{1\}, G/N$, then by the correspondence theorem $M = \pi^{-1}(\bar{M})$ would be a normal subgroup of G with $N \subsetneq M \subsetneq G$, contradicting the maximality of $|N|$. By induction N has a composition series, and appending G to it gives a composition series of G.

Example. Let us write out the composition series for $\mathbb{Z}/6\mathbb{Z}$. Indeed,

$$\mathbb{Z}/6\mathbb{Z} \rhd \langle 3 \rangle \rhd \{0\} \quad \text{with factors } \mathbb{Z}/3\mathbb{Z},\ \mathbb{Z}/2\mathbb{Z},$$
$$\mathbb{Z}/6\mathbb{Z} \rhd \langle 2 \rangle \rhd \{0\} \quad \text{with factors } \mathbb{Z}/2\mathbb{Z},\ \mathbb{Z}/3\mathbb{Z}.$$

Remark: For finite groups, composition series exist, but are not unique! But the composition factors are unique up to isomorphism and reordering, this is the Jordan-Hölder theorem.

Definition. A group G is **solvable** if there exists a sequence

$$\{1\} = G_0 \lhd G_1 \lhd \cdots \lhd G_k = G$$

such that G_i/G_{i-1} is abelian for each i.

Homework: We will prove that a finite group G is solvable if and only if all composition factors are abelian!

Proposition.

1. If G is solvable, then any subgroup of G and any quotient of G is also solvable.
2. If $N \lhd G$ and both N and G/N are solvable, then so is G.

Proof. (1) is left for our homework. To prove (2), we use the definition. To this end, since N is solvable, we have a sequence

$$\{1\} = N_0 \lhd N_1 \lhd \cdots \lhd N_k = N$$

such that N_i/N_{i-1} is abelian for every i. Since G/N is solvable, we have a sequence

$$\{1\} = \overline{G_0} \lhd \overline{G_1} \lhd \cdots \lhd \overline{G_\ell} = G/N$$

such that $\overline{G_j}/\overline{G_{j-1}}$ is abelian for every j. Now, let $\pi : G \to G/N$ be the quotient map and set $G_j = \pi^{-1}(\overline{G_j})$. By the correspondence theorem (the 4th isomorphism theorem), $G_0 = \pi^{-1}(\{1\}) = N$ and $G_\ell = G$, and part (2) of that theorem gives $G_{j-1} \lhd G_j$ and

$$G_j/G_{j-1} \cong \overline{G_j}/\overline{G_{j-1}},$$

which is abelian. Concatenating the two sequences, we obtain

$$\{1\} = N_0 \lhd \cdots \lhd N_k = N = G_0 \lhd G_1 \lhd \cdots \lhd G_\ell = G,$$

in which every successive quotient is abelian. Hence G is solvable. □

Example. Any abelian group G is solvable, via $\{1\} \lhd G$. Moreover, D_{2n} is also solvable, since $\{1\} \lhd \langle r \rangle \lhd D_{2n}$ works.

Non examples include any group that is nonabelian and simple, for instance A_n for $n \geq 5$ (indeed, a sequence with abelian quotients would have to be $\{1\} \lhd G$, forcing G to be abelian).

Now, we shift gears back to group actions. Recall that $G \curvearrowright A$ is the same as having a permutation representation

$$G \to S_A, \qquad g \mapsto \varphi_g, \quad \text{where } \varphi_g(a) = g \cdot a,$$

which is a homomorphism as discussed earlier in the course. Recall also that $\mathrm{Stab}_G(a) = \{g \in G : g \cdot a = a\} \leq G$. Group actions induce an equivalence relation (we will talk about orbits in tutorial). The equivalence relation that we are interested in is the following:

$$a \sim b \iff \exists g \in G \text{ such that } a = g \cdot b.$$

In a nutshell, the equivalence classes of this relation are called **orbits**, so they form a partition of the set A:

$$A = \bigsqcup \text{orbits}.$$

Notation: we denote the orbit of a by $G \cdot a = \{g \cdot a : g \in G\}$. (Do not confuse this with cosets!)

Definition. A group action is said to be **transitive** if there exists only one orbit.

If $H \leq G$, we denote

$$G/H = \text{the set of all left cosets } gH,$$
$$H\backslash G = \text{the set of all right cosets } Hg.$$

Theorem. (Orbit-Stabilizer Theorem) For any $a \in A$, there exists a bijection

$$G/\mathrm{Stab}_G(a) \to G \cdot a, \qquad g\,\mathrm{Stab}_G(a) \mapsto g \cdot a.$$

In particular, $|G \cdot a| = (G : \mathrm{Stab}_G(a))$, and if G is finite, then

$$|G \cdot a| = \frac{|G|}{|\mathrm{Stab}_G(a)|}.$$

Proof. This is an ideal theorem, in the sense that the proof is easy but this result is useful. In the interest of my sanity, let $S = \mathrm{Stab}_G(a)$ and consider the map $\varphi : G/S \to G \cdot a$ given by $gS \mapsto g \cdot a$. For $g, h \in G$ we have

$$gS = hS \iff h^{-1}g \in S \iff h^{-1}g \cdot a = a \iff g \cdot a = h \cdot a.$$

The implication ($\Rightarrow$) shows that φ is well defined, and the implication ($\Leftarrow$) shows that φ is injective. Moreover, surjectivity follows by definition of the orbit. Hence φ is a bijection, so $|G \cdot a| = |G/S| = (G : S)$, and when G is finite this equals $|G|/|S|$ by Lagrange's theorem. □

Example. We look at the rotational symmetries of a dodecahedron. To this end, we pick a suitable action, say $G \curvearrowright \{12 \text{ faces}\}$. Note that we have only one orbit, the order of the stabilizer of a face is 5 (the rotations about the axis through its center), and the order of G is 60, so by the orbit-stabilizer theorem, we have that

$$|G \cdot a| = \frac{60}{5} = 12,$$

which is indeed the number of faces (equivalently, $|G| = 12 \cdot 5 = 60$).

More Group Actions and Simplicity of A_n

We talked a bit more about group actions, $\mathrm{Stab}_G(a) \leq G$, and orbits ($G \cdot a = \{g \cdot a : g \in G\}$), do not confuse these with cosets!

Most importantly, we proved the orbit-stabilizer theorem, which gives a bijection

$$G/\mathrm{Stab}_G(a) \to G \cdot a, \qquad g \cdot \mathrm{Stab}_G(a) \mapsto g \cdot a$$

where $G/\mathrm{Stab}_G(a)$ is the set of left cosets. Moreover (for G finite), $|G| = |\mathrm{Stab}_G(a)| \cdot |G \cdot a|$.
Remark: Burnside's lemma gives a formula for the number of orbits. See Dummit and Foote.

Example. $G = S_n \curvearrowright \{1, 2, \ldots, n\}$. Recall that the action is transitive if there exists only one orbit.

Say $\sigma = (1\ 2\ 3\ 4)(5\ 6)(7)$ for $n = 7$. The cyclic subgroup $\langle\sigma\rangle \curvearrowright \{1, \ldots, n\}$, and the orbits of this action are the cycles occurring in the disjoint cycle decomposition (here $\{1, 2, 3, 4\}$, $\{5, 6\}$ and $\{7\}$).

Theorem. **(Cayley's Theorem)** Any group G is isomorphic to a subgroup of the symmetric group S_G. (In particular, if $|G| = n$, then G is isomorphic to a subgroup of S_n.)

Proof. We harness the power of the permutation representation $G \to S_A$. Let $A = G$, with the left translation action $g \cdot a = ga$. So we have

$$\varphi : G \to S_G, \qquad g \mapsto (a \mapsto g \cdot a).$$

Claim: $\ker\varphi = \{1\}$. Indeed, if $\varphi(g) = 1$, then $g \cdot a = a$ for all $a \in G$, and in particular $g = g \cdot 1 = 1$.

Hence, due to the first isomorphism theorem, $G \cong \mathrm{im}\,\varphi \leq S_G$. □

Remark: If $G \curvearrowright G$ by conjugation, then $\ker\varphi = Z(G)$.

Not useful in practice, say we want to classify all groups of order 20, S_{20} is crazy!

Remark: Any subgroup $H \leq G$ is a stabilizer (in some action of G). Indeed, consider

$$G \curvearrowright G/H = \{gH : g \in G\}, \qquad g \cdot xH = gxH.$$

We compute $\mathrm{Stab}_G(H)$. Indeed, $gH = H$ if and only if $g \in H$, so $\mathrm{Stab}_G(H) = H$.

We talk more about conjugation, indeed, if $G \curvearrowright G$ via $g \cdot a = gag^{-1}$, recall that $\mathrm{Stab}_G(a) = C_G(a)$ (centralizer).

Definition. The orbits of the conjugation action are called the **conjugacy classes**. We say that $a, b \in G$ are **conjugate** if they lie in the same conjugacy class.

Remark: We write $\mathrm{ccl}_G(a)$ to denote the conjugacy class of a.

Now, we can discuss a numerical version of the orbit-stabilizer theorem, indeed

$$|\mathrm{ccl}_G(a)| = (G : C_G(a)).$$

Theorem. **(Class Equation)** Let G be a finite group. Then

$$|G| = |Z(G)| + \sum_{i=1}^{r} (G : C_G(g_i)),$$

where $g_1, \ldots, g_r$ are representatives of the conjugacy classes of size > 1.

Proof sketch. The idea is that

$$G = \bigsqcup \text{conjugacy classes},$$

namely a disjoint union of conjugacy classes. Moreover, $g \in Z(G)$ if and only if $\operatorname{ccl}(g) = \{g\}$. Furthermore, $|\operatorname{ccl}(g_i)| = (G : C_G(g_i))$ by the orbit-stabilizer theorem. Hence

$$|G| = |Z(G)| + \sum_{i=1}^{r} |\operatorname{ccl}(g_i)| = |Z(G)| + \sum_{i=1}^{r} (G : C_G(g_i)). \qquad \square$$

Remark: Similar matrices are a special case of conjugacy classes (conjugation in $\mathrm{GL}_n(k)$).

Lemma. If $|G| = p^n$ (p prime, $n \geq 1$) then $Z(G) \neq \{1\}$.

Remark: Such a group is called a p-group.

Proof. By the class equation,

$$\underbrace{|G|}_{p^n} = |Z(G)| + \sum_{i=1}^{r} (G : C_G(g_i)),$$

where each index $(G : C_G(g_i))$ is > 1 and divides $p^n = |G|$, so it is divisible by p. So $p \mid |Z(G)|$ and $Z(G) \neq \{1\}$. $\square$

Remark: We could say

$$\underbrace{|G|}_{p^n} = \sum |\operatorname{ccl}| \equiv \#\{\text{conjugacy classes of size } 1\} \pmod{p},$$

and $|G| = p^n \equiv 0 \pmod{p}$.

Corollary. Any group of order p^2 is abelian. In fact, it is isomorphic to one of

1. $\mathbb{Z}/p^2\mathbb{Z}$
2. $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$

Proof. By the lemma, $Z(G) \neq \{1\}$, so $G/Z(G)$ has order p or 1 by Lagrange. If $G/Z(G) = \{1\}$, then $Z(G) = G$, so G is abelian. And if $|G/Z(G)| = p$, then $G/Z(G)$ is isomorphic to $\mathbb{Z}/p\mathbb{Z}$, which is cyclic. Moreover, $G/Z(G)$ cyclic implies G is abelian (worksheet/exercise!).

For the last part, see the book. $\square$

Remark: There exist non-abelian groups of order p^3, say Q_8, D_8 ($p = 2$).

Let us discuss conjugacy in S_n. In tutorial, we discussed how we can efficiently conjugate cycles in S_n. More precisely, if $\sigma \in S_n$, then

$$\sigma \underbrace{(i_1 \cdots i_k)}_{k\text{-cycle}} \sigma^{-1} = \underbrace{(\sigma(i_1)\ \sigma(i_2) \cdots \sigma(i_k))}_{\text{again, a } k\text{-cycle}}$$

Since conjugation is an automorphism, we can apply this cycle by cycle. For example

$$\sigma(1\ 2\ 3)(4\ 5)\sigma^{-1} = \sigma(1\ 2\ 3)\sigma^{-1}\cdot\sigma(4\ 5)\sigma^{-1} = \underbrace{(\sigma(1)\ \sigma(2)\ \sigma(3))(\sigma(4)\ \sigma(5))}_{\text{type } 3,2} \qquad (*)$$

So conjugation preserves the cycle type.

Proposition. $\tau_1, \tau_2 \in S_n$ are conjugate if and only if τ_1, τ_2 have the same cycle type.

Proof. ($\Rightarrow$) See the above discussion.

($\Leftarrow$) Conversely, we write the disjoint cycle decomposition, including cycles of length 1:

$$\tau_1 = (i_1 \cdots i_k)(i_1' \cdots i_{k'}')(i_1'' \cdots)\cdots$$
$$\tau_2 = (j_1 \cdots j_k)(j_1' \cdots j_{k'}')(j_1'' \cdots)\cdots$$

Each has the same cycle type (note that each of $1, 2, \ldots, n$ appears exactly once). Define σ by $\sigma(i_\ell) = j_\ell$, $\sigma(i_\ell') = j_\ell'$, $\ldots$ So that $\sigma\tau_1\sigma^{-1} = \tau_2$ by $(*)$. □

Therefore,

$$\begin{aligned} \#\text{ of conjugacy classes in } S_n &= \#\text{ of cycle types} \\ &= \#\text{ of partitions of } n. \end{aligned}$$

Here $n = n_1 + \cdots + n_r$ with integers $n_1 \geq n_2 \geq \cdots \geq n_r > 0$ $(r \geq 1)$.

Example.

Cycle type	Partition
$(1\ 2\ 3\ 4\ 5)$	5
$(1\ 2\ 3\ 4)$	$4+1$
$(1\ 2\ 3)(4\ 5)$	$3+2$
$(1\ 2\ 3)$	$3+1+1$
$(1\ 2)(3\ 4)$	$2+2+1$
$(1\ 2)$	$2+1+1+1$
1	$1+1+1+1+1$

So there are 7 conjugacy classes in S_5.

We now discuss the simplicity of A_n, this is not easy and holds for some n $(n \geq 5)$.

Theorem. A_n is simple for all $n \geq 5$.

Remark: The book has a different proof for $n = 5$.

Proof. Let $1 \neq N \lhd A_n$. It suffices to show that $N = A_n$.

Lemma. Every $\sigma \in A_n$ is a product of 3-cycles. Namely A_n is generated by 3-cycles.

Proof of lemma. σ is a product of 2-cycles, an even number of 2-cycles as $\sigma \in A_n$. So, it suffices to show that the product of two 2-cycles is a product of 3-cycles. We have 3 possibilities (up to relabeling):

$$\begin{aligned}(1\ 2)(1\ 2) &= 1\\ (1\ 2)(2\ 3) &= (1\ 2\ 3)\\ (1\ 2)(3\ 4) &= (1\ 2)(2\ 3)(2\ 3)(3\ 4)\\ &= (1\ 2\ 3)(2\ 3\ 4)\end{aligned}$$

□

We prove another lemma.

Lemma. If N contains a 3-cycle, then $N = A_n$.

Proof of lemma. WLOG $(1\ 2\ 3) \in N$. We make the following claim.

Claim: $\forall \sigma \in S_n$, $\sigma(1\ 2\ 3)\sigma^{-1} \in N$.

Proof of claim: If $\sigma \in A_n$, then we are done as $N \triangleleft A_n$. Now, if $\sigma \notin A_n$, we write $\sigma = \tau(4\ 5)$, where τ is even, so it is in A_n. Then

$$\begin{aligned}\sigma(1\ 2\ 3)\sigma^{-1} &= \tau \underbrace{(4\ 5)(1\ 2\ 3)(4\ 5)^{-1}}_{(1\ 2\ 3)} \tau^{-1}\\ &= \tau(1\ 2\ 3)\tau^{-1} \in N\end{aligned}$$

as $\tau \in A_n$.

By the proposition proved earlier (all 3-cycles are conjugate in S_n), N contains all 3-cycles, thus by Lemma 1, $N = A_n$. □

Now, we have 4 cases (these exhaust all possibilities for a nontrivial $\sigma \in N$).

Case 1: N contains σ that has a cycle of length ≥ 4 (in the disjoint cycle decomposition). Now WLOG, $\sigma = (1\ 2\ 3\ \cdots\ k)\tau$, where $k \geq 4$ and τ is a product of disjoint cycles. Then we have

$$\underbrace{\underbrace{(1\ 2\ 3)\sigma(1\ 2\ 3)^{-1}}_{\in N} \cdot \underbrace{\sigma^{-1}}_{\in N}}_{\in N} = (2\ 3\ 1\ \cdots\ k)\tau\tau^{-1}(k\ k{-}1\ \cdots\ 1) = (1\ 2\ 4).$$

By Lemma 2 we have $N = A_n$.

Case 2: N contains σ with at least two 3-cycles (and no longer cycles by Case 1). Indeed, WLOG $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$, where τ is a product of disjoint cycles. Then

$$(1\ 2\ 4)\sigma(1\ 2\ 4)^{-1}\sigma^{-1} = (2\ 4\ 3)(1\ 5\ 6)\tau \cdot \tau^{-1}(6\ 5\ 4)(3\ 2\ 1) = (1\ 2\ 5\ 3\ 4) \in N.$$

This is a 5-cycle, so we are in Case 1, and $N = A_n$.

Case 3: N contains σ with exactly one 3-cycle (and 2-cycles, 1-cycles). WLOG $\sigma = (1\ 2\ 3)\tau$, where τ is a product of disjoint 2-cycles. Then

$$\sigma^2 = (1\ 2\ 3)^2 \underbrace{\tau^2}_{=1} = (1\ 3\ 2) \in N,$$

and we are done by Lemma 2.

Case 4: N contains σ which is a product of ≥ 2 (even!) 2-cycles. WLOG (last time I promise) $\sigma = (1\ 2)(3\ 4)\tau$, where τ is a product of disjoint cycles. Then

$$\big((1\ 2\ 3)\sigma(1\ 2\ 3)^{-1}\big)\sigma^{-1} = (2\ 3)(1\ 4)\tau \cdot \tau^{-1}(4\ 3)(2\ 1) = (1\ 3)(2\ 4) \in N.$$

Finally (this is horrible),

$$(1\ 3)(2\ 4)\Big(\underbrace{(1\ 3\ 5)(1\ 3)(2\ 4)(1\ 3\ 5)^{-1}}_{(3\ 5)(2\ 4)}\Big) = (1\ 3\ 5) \in N.$$

Again... by Lemma 2, $N = A_n$. This finishes the proof that A_n is simple for $n \geq 5$.

□

Remark: A_4 is not simple.

The Sylow Theorems

Last time, we proved that A_n is simple for $n \geq 5$. Today we will define what an automorphism is (we saw it in the previous homework but we'll add more detail now), and then introduce one of the most important theorems in finite group theory, known as Sylow's theorems. We will try to prove one of them, time permitting.

Definition. An **automorphism** is an isomorphism $G \to G$. The set

$$\mathrm{Aut}(G) = \{\varphi : G \to G : \varphi \text{ is an isomorphism}\}$$

is a group under composition.

Lemma. If $N \triangleleft G$, there is a homomorphism

$$\begin{aligned} \varphi : G &\to \mathrm{Aut}(N) \\ g &\mapsto \left(n \mapsto gng^{-1}\right) \end{aligned}$$

with $\ker \varphi = C_G(N)$.

Remark: This generalizes $c_x : G \to G$ via $g \mapsto xgx^{-1}$.

Corollary. $G/C_G(N)$ is isomorphic to a subgroup of $\mathrm{Aut}(N)$. In particular, if $N = G$, then $G/Z(G) \cong \mathrm{Inn}(G)$, namely the inner automorphisms.

Example. If G is abelian, then $\mathrm{Inn}(G) = \{1\}$.

Proof of Lemma. First, for any $g \in G$, $\varphi(g) \in \mathrm{Aut}(N)$ as $N \triangleleft G$, and it has inverse $\varphi(g^{-1})$. Then we claim that φ is a homomorphism. To this end, note that

$$\begin{aligned} \varphi(gh)(n) &= ghnh^{-1}g^{-1}, \\ \varphi(g)\big(\varphi(h)(n)\big) &= g\left(hnh^{-1}\right)g^{-1}. \end{aligned}$$

Hence φ is a group homomorphism. Finally, $\ker \varphi = C_G(N)$ by definition. □

Example. $A_n \triangleleft S_n$ $(n \geq 4)$. Take $\sigma \in S_n \smallsetminus A_n$, say $\sigma = (12)$. Then the map $A_n \to A_n$, $\tau \mapsto \sigma\tau\sigma^{-1}$, is not an inner automorphism of A_n.

Indeed, the kernel of the homomorphism $S_n \to \mathrm{Aut}(A_n)$ is $C_{S_n}(A_n) = 1$ (exercise!), so this map is injective. This is an automorphism that is not in $\mathrm{Inn}(A_n)$.

Example. One can show that every automorphism of S_n is inner when $n \neq 6$. But S_6 has a non-inner automorphism (see textbook).

We shift gears towards the Sylow theorems. Indeed, given p prime, $|G| = p^a m$, $p \nmid m$.

Definition. A **Sylow p-subgroup** is a subgroup of G of order p^a.

Remark: It is a largest p-subgroup, by Lagrange's theorem.

Example. $G = S_3$. We write down the $p = 2$ and $p = 3$ Sylow subgroups. Note that $|S_3| = 6$. We have

$$\begin{aligned} p = 2: &\quad \langle (12) \rangle,\ \langle (23) \rangle,\ \langle (13) \rangle \\ p = 3: &\quad \langle (123) \rangle \end{aligned}$$

So there are 3 such 2-Sylow subgroups and 1 such 3-Sylow subgroup.

Notation: We denote by $\mathrm{Syl}_p(G) = \{\text{Sylow } p\text{-subgroups}\}$ and $n_p = |\mathrm{Syl}_p(G)|$.

Theorem. **(Sylow)** Fix p prime and let G be a group of order $p^a m$, $p \nmid m$, $a \geq 0$.

(1) G has a Sylow p-subgroup, namely $n_p > 0$.

(2) Any two Sylow p-subgroups are conjugate (under G). (Even more: if P is a Sylow p-subgroup and Q is any p-subgroup, then $Q \leq gPg^{-1}$ for some g.)

(3) $n_p \mid m$ and $n_p \equiv 1 \pmod{p}$.

Corollary. If $n_p = 1$, then the unique Sylow p-subgroup is normal in G. Namely, $gPg^{-1} = P$ for all $g \in G$ (as $n_p = 1$).

Proof of the first theorem. (Warning: this is tricky!) We use a group action. Indeed, define

$$S = \{\text{subsets } X \text{ of } G,\ |X| = p^a\}.$$

Let G act on S by translation, $g \cdot X = gX$. The proof is tricky/long, so I will partition it claim by claim.

Claim 1: $|S| \not\equiv 0 \pmod{p}$.

Proof of Claim 1. We summon combinatorics. We show that

$$|S| = \binom{|G|}{p^a} = \binom{p^a m}{p^a} \equiv m \not\equiv 0 \pmod{p}.$$

Indeed, we use a slick proof. We use the Freshman's dream. Namely, $(x+y)^p = x^p + y^p$ in $\mathbb{F}_p[x,y]$ (as $p \mid \binom{p}{i}$ for $0 < i < p$). Then by induction, $(x+y)^{p^k} = x^{p^k} + y^{p^k}$. Then

$$(x+y)^{p^a m} = \left(x^{p^a} + y^{p^a}\right)^m = x^{p^a m} + \binom{m}{1} x^{p^a (m-1)} y^{p^a} + \cdots$$

Comparing the coefficient of $x^{p^a(m-1)} y^{p^a}$ on both sides (the left side has coefficient $\binom{p^a m}{p^a}$),

$$\binom{p^a m}{p^a} \equiv \binom{m}{1} = m \quad \text{in } \mathbb{F}_p.$$

Namely,

$$|S| = \binom{p^a m}{p^a} \equiv m \not\equiv 0 \pmod{p}.$$

Recall that orbits partition S, so there is an orbit of some $X \in S$ of size $\not\equiv 0 \pmod{p}$. So, by the orbit-stabilizer theorem, $|G| = |\mathrm{Orb}(X)| \cdot |\mathrm{Stab}_G(X)|$ with $p \nmid |\mathrm{Orb}(X)|$, and hence

$$|\mathrm{Stab}_G(X)| \equiv 0 \pmod{p^a}. \tag{$*$}$$

This proves Claim 1.

Claim 2: $|\mathrm{Stab}_G(X)| \leq p^a$. (This implies by $(*)$ that $|\mathrm{Stab}_G(X)| = p^a$, so $\mathrm{Stab}_G(X)$ is a p-Sylow subgroup!)

Proof of Claim 2. Pick $x \in X$ $(\subseteq G)$ and define

$$\begin{aligned} f : \mathrm{Stab}_G(X) &\to X \\ g &\mapsto gx \in X \end{aligned}$$

(not a homomorphism). It is injective by the cancellation law. Hence $|\mathrm{Stab}_G(X)| \leq |X| = p^a$. □